Model Optimization and Tuning Phase Template

Date	25 February 2026
Team ID	259422
Project Title	Online Payments Fraud Detection using Machine Learning
Maximum Marks	10 Marks

Model Optimization and Tuning Phase

The objective of hyperparameter tuning was to:

- Improve fraud detection recall
- Increase F1 Score
- Reduce false negatives
- Improve generalization capability
- Prevent overfitting

Tuning Method Used:

- Scoring Metric: **Recall (Fraud Class)**

Recall was prioritized because undetected fraud (False Negatives) results in financial loss.

Hyperparameter Tuning Documentation (6 Marks):

Model	Tuned Hyperparameters	Optimal Values
Decision Tree Classifier	`max_depth`, `min_samples_split`, `min_samples_leaf`, `criterion`	`max_depth=12`, `min_samples_split=10`, `min_samples_leaf=4`, `criterion='gini'`
Random Forest Classifier	`n_estimators`, `max_depth`, `min_samples_split`, `class_weight`	`n_estimators=200`, `max_depth=None`, `min_samples_split=5`, `class_weight='balanced'`
Extra Trees Classifier	`n_estimators`, `max_depth`, `min_samples_split`, `min_samples_leaf`, `class_weight`	`n_estimators=300`, `max_depth=None`, `min_samples_split=5`, `min_samples_leaf=2`, `class_weight='balanced'`

Performance Metrics Comparison Report (2 Marks):

Model	Baseline Metric (Recall)	Optimized Metric (Recall)
Decision Tree Classifier	0.78	0.89
Random Forest Classifier	0.70	0.79
Extra Trees Classifier	0.69	0.78

Final Model Selection Justification (2 Marks):

Final Model	Reasoning
Model 1 (Decision Tree Classifier)	Because of imbalanced data class (very loow number of fraud transactions against very high number of legit transactions), accuracy is high for all the models. Of all the 4 Classification ML Models used above, Decision Tree Classifier turns out to be the most accurate predictor for the input dataset. Since in bank transactions, missing any Fraud transaction and clearing it as Legit is the biggest business risk, keeping the ratio of False Negatives (Recall value) should be our main focus. Decision Tree is a clear winner with the highest recall & F-1 Score.